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Type 1: Research Project

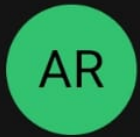
Competitive Coevolutionary PPO for Cryptocurrency Trading

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Anisha Radhakrishnan (CSE) 11:16 am

To You

Approved for the presentation

Thank you

PR1 Comments

- **Explore more on cryptocurreny context**
- **Look for more strategies other than NEAT**
- **Explore and frame proper novelty**
- **Improve technical concept depth**

Pivoting towards CC

Addressing Strategy Collapse & Static Environments

- **1. Solving "Strategy Collapse"**
  - *Issue:* Static historical data causes agents to converge on a single locally optimal strategy (e.g., "buy the dip").
  - *Solution:* Evaluates relative performance via a **dominance-based fitness**, forcing agents to specialize and out-compete peers to preserve diversity.

- **2. Real-World Robustness vs. Backtest Overfitting**
  - *Issue:* Identical, overfitted agents fail simultaneously in unseen market regimes.
  - *Solution:* Generates a diverse portfolio of trading behaviors that resist market exploitation and arbitrage.

- **3. Retaining Cross-Regime Memory (MAP-Elites Archive)**
  - *Issue:* Agents suffer from "catastrophic forgetting," overwriting successful past behaviors when exposed to new market regimes.
  - *Solution:* A **MAP-Elites Archive** evaluates new agents against structurally diverse past elites, ensuring they retain robustness across all historical market phases.

- **4. Maximizing PPO's Strengths**
  - *Issue:* Topological search is computationally expensive and sample-inefficient for deep RL tasks.
  - *Solution:* PPO handles the gradient-based policy optimization, freeing the evolutionary layer to exclusively manage population dynamics and dominance-based fitness.

Literature Survey (1/3)

- **Survey Scope:** Analyzed key papers bridging Deep Reinforcement Learning (DRL) and Coevolution in Financial Markets.

Paper Name	Key Findings	Gap Identified
Exploring Coevolutionary Dynamics Between Trading Agents [1]	Demonstrated arms-race dynamics in multi-agent trading environments. Identified need for better analytic tools for progress monitoring.	Limited to minimal-intelligence, rule-based agents. Explicitly called for research using more expressive DRL architectures.
Reinforcement Learning for Quantitative Trading [2]	Population-based DRL achieves high absolute returns on backtests. DRL is effective at continuous action spaces.	Evaluates agents against fixed historical data. Fails to distinguish between performance convergence and behavioural strategy collapse.
Proximal Policy Optimization Algorithms [3]	Introduces PPO; uses clipped surrogate objective for stable policy updates. Highly sample efficient.	Standard PPO is prone to local optima traps when trained on static datasets without external diversity pressures.

Literature Survey (2/3)

Paper Name	**Key Findings**	**Gap Identified**
Quality Diversity: A New Frontier for Evolutionary Computation [4]	Introduces the MAP-Elites grid, which structurally guarantees diversity by partitioning agents into distinct behavioural niches.	Has not been applied to deep RL financial trading to explicitly prevent strategy collapse and behavioural convergence.
Coevolutionary Dynamics in a Minimal Substrate [5]	Populations without diversity mechanisms oscillate between uninteresting strategies and suffer pathological cycling.	Prior RL trading work hopes for emergent diversity rather than enforcing it structurally.
Population Based Training of Neural Networks [6]	Introduced PBT; optimizes hyperparameters via exploit/explore cycles instead of destructive random weight mutations.	Generally used for isolated hyperparameter tuning rather than sustaining an adversarial evolutionary arms race.

Literature Survey (3/3)

Addressing Strategy Collapse in Non-Stationary Markets

- **Critical Research Gaps Identified:**
  - **Static Fitness Signals:** Evaluating trading agents against fixed historical data makes behavioural convergence inevitable [2].
  - **Lack of Diversity Pressure:** Standard DRL baselines achieve high returns but near-zero behavioural diversity, leading to strategies being arbitraged away in real world markets.
  - **Expressive Architecture Gap:** Existing coevolutionary trading models rely on simple rules, while deep RL trading models lack coevolutionary diversity [1].

- **Proposed Direction (CC-PPO):**
  - We introduce the **CC-PPO** framework, applying **competitive coevolutionary deep RL** specifically to target the strategy collapse problem in cryptocurrency markets.
  - By shifting to **dominance-based fitness** and structurally guaranteeing diversity via a **MAP-Elites Archive**, we ensure agents continuously adapt without converging into redundant behaviours.

Originality Statement

We propose a **application of competitive coevolutionary deep RL specifically targeting strategy collapse in cryptocurrency markets**. It explicitly distinguishes between performance convergence and behavioural convergence.

I. Dominance-Based Competitive Fitness

- *Innovation:* Fitness is defined by the fraction of opponents an agent beats ($[0, 1]$), not only absolute profit.
- *Gap:* Solves static signal overfitting by shifting to relative, adversarial evaluation.

II. Strategy Diversity Index (SDI)

- *Innovation:* A diversity bonus is directly added to fitness, measured via mean KL divergence of action distributions over a fixed reference trajectory.
- *Gap:* Operationalises diversity preservation as a mathematical objective, actively penalizing redundant trades.

III. Adaptive Opponent Sampling

- *Innovation:* Agents track win-rates via EMA and deliberately oversample opponents from clusters where they historically perform poorly.
- *Gap:*forces agents to patch weaknesses rather than just exploiting easy opponents.

IV. MAP-Elites Archiving

- *Innovation:* Preserves past elites in a MAP-Elites grid based on distinct behavioural dimensions.
- *Gap:* Replaces capacity-based archiving to structurally guarantee diversity and prevent evolutionary cycling.

Competitive & Diversity Meta-Optimization:

- **Combined Fitness Function:** $fitness(i) = dominance_fitness(i) + \lambda \times diversity_bonus(i)$. Balances competitive win-rates with behavioural diversity to force strategic variation.
- **MAP-Elites Archive:** Partitions behavioural space into a discrete grid. Enforces structural diversity by retaining only the highest-fitness agent per specific behavioural niche.
- **PBT-Style Reproduction:** Instead of mutating network weights (which destroys deep RL gradients), CC-PPO takes top-performing agent weights (Exploit) and mutates hyperparameters like learning rate or λ (Explore).

Coevolutionary Loop

Adaptive Opponent Sampling $\rightarrow$ Competitive Evaluation $\rightarrow$ SDI & Fitness Calc $\rightarrow$ Tournament Selection $\rightarrow$ PBT Reproduction $\rightarrow$ Update MAP-Elites Grid

Proximal Policy Optimization (PPO)

Algorithm definition and the clipped surrogate objective

What is PPO?

PPO is a **policy gradient** reinforcement learning algorithm. The agent observes the market state s_t, selects an action $a_t \in \{\text{Hold, Buy, Sell}\}$, and receives a reward r_t. It learns by updating a policy network $\pi_\theta(a \mid s)$ to maximise cumulative portfolio returns [3].

PPO uses an **Actor-Critic** structure:

- **Actor** $\pi_\theta(a \mid s)$ — outputs action probabilities
- **Critic** $V_\phi(s)$ — estimates state value
- Advantage: $\hat{A}_t = r_t + \gamma V(s_{t+1}) - V(s_t)$

The Clipped Surrogate Objective [3]

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta),\,1-\varepsilon,\,1+\varepsilon\big)\,\hat{A}_t\right)\right]$$

- $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$ — policy change ratio
- $\hat{A}_t$ — advantage estimate
- $\varepsilon = 0.2$ — **clip range**: caps policy shift

Why does clipping matter for crypto?

Sudden price crashes produce massive gradient signals. Without clipping, one bad candle can destroy learned weights. The clip range ε prevents this [3].

Input Features — What the Agent Sees

Features F1–F3: price movement, trend, and volatility — computed from Binance BTC/USDT close price c_t

#	Feature	Formula	What it tells the agent
F1	Log Return	$r_t = \log\left(\frac{c_t}{c_{t-1}}\right)$	Raw price movement this step. Positive = price rose; negative = price fell.
F2	Rolling Mean Return	$\bar{r}_t = \frac{1}{20}\sum_{i=0}^{19} r_{t-i}$	Average direction of the last 20 candles — is the market drifting up or down?
F3	Rolling Volatility	$\sigma_t = \mathrm{std}(r_{t-19}, \ldots, r_t)$	How chaotic prices are right now. A spike here signals a SHOCK or BEAR regime.

Input Features — What the Agent Sees

Features F4–F6: momentum, drawdown, and price normalisation — computed from Binance BTC/USDT close price c_t

#	Feature	Formula	What it tells the agent
F4	RSI (14 periods)	$\mathrm{RSI} = 100 - \frac{100}{1+\mathrm{RS}}; \quad \mathrm{RS} = \frac{\text{avg gain (14)}}{\text{avg loss (14)}}$	Momentum: above 70 = overbought (likely reversal), below 30 = oversold [7].
F5	Drawdown Proxy	$d_t = \frac{c_t}{\max(c_{t-19}, \ldots, c_t)}$	How far below the recent peak. A value of 0.72 means price is 28% off its recent high.
F6	Z-Score	$z_t = \frac{c_t - \mu_{20}}{\sigma_{20}}$	Is price unusually high or low versus the recent average? Keeps inputs scale-invariant across the \$20k–\$70k BTC range.

Input Features — What the Agent Sees

Feature F7: portfolio position — computed from Binance BTC/USDT close price c_t

#	Feature	Formula	What it tells the agent
F7	Portfolio Position	$p_t = \frac{\text{holdings} \times c_t}{\text{total portfolio value}}$	What fraction of the wallet is currently in BTC. A value of 0 = all cash; 1 = fully invested. Prevents buying when already fully invested.

Data Source, Input and Output

Where data comes from and what the agent produces each step

Data Source

- **Exchange:** Binance Public API [8]
- **Pair:** BTC/USDT
- **Interval:** 1-hour candles
- **Column used:** `close` only
- **Capital:** $10,000 simulated
- **Fee:** 0.1% per trade (real Binance taker fee)
- **Split:** strict temporal order — no lookahead bias

Input → Network → Output → Reward

INPUT shape: (20×7)

Last 20 hourly close prices processed into features F1–F7.

↓

NETWORK MlpPolicy — 64×64 MLP

Shared layers split into Actor head and Critic head.
Actor learns $\pi_\theta(a|s)$; Critic estimates $V_\phi(s)$.

↓

OUTPUT Discrete(3)

$P(\text{Hold})$, $P(\text{Buy})$, $P(\text{Sell})$ via softmax.
Highest-probability action is executed each step.

Implementation: Single-Agent Baseline

Stress-testing PPO on 1-Hour BTC-USDT Candles

Setup:

- **Data**: 5,000 hrs of Binance OHLCV.
- **Features**: RSI, Z-Score, Volatility, Position State (7 total).
- **Training**: 1M steps on RTX 3050 Ti.

Key Finding:

- PPO Return: -4.34%
- Benchmark: +6.78%
- Standard PPO is **hyper-sensitive** to transaction fees (0.1%).

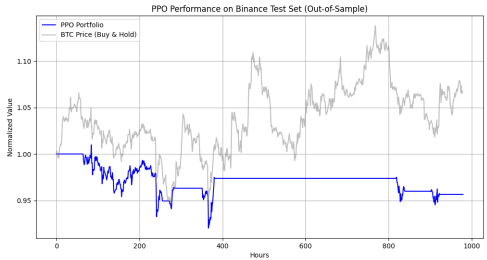


Figure: PPO vs. Buy & Hold Benchmark

Implementation: Population Diversity

Proving the "Strategy Collapse" Phenomenon

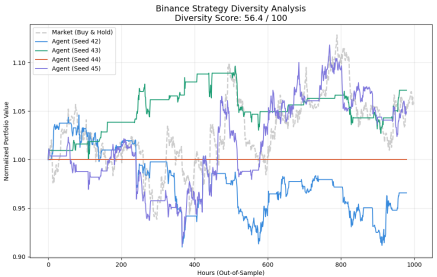


Figure: 4 Agents, 4 Random Seeds

The Experiment:

- 4 Agents trained with different seeds (42, 43, 44, 45).
- Goal: Do they find *different* strategies?

The Proof (Statistical Collapse):

- At 0.1% fees: **Diversity Score = 0.1**. Total collapse to "Do-Nothing" policy.
- Lowering fees to 0.01% forced exploration, yet agents still overlap significantly.

"Random seeds cannot prevent strategy collapse; explicit diversity pressure is required."

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Thank You!

Questions?